

FIRAT HAJIYEV

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EDUCATION

University of Central Florida, Orlando

2024 - May 2028

Bachelor of Science, Computer Science (GPA: 3.84)

High School No 194, Baku

Jun 2024

High School Diploma

• **Achievements:** Honors Diploma of High School

SKILLS

• **Skills:** Software Development, Embedded Software Development, C++, C, Python, Bash, FreeRTOS, Driver Development, Memory Management, ISRs, Kalman Filtering, Git/Github, Edge AI, AI/ML, Embedded Linux, Zynq UltraScale+ MPSoC, Data Structures, Signal Processing, Java, FPGA Development, Scrum, U-Boot

Projects

AVR-Zero-Alloc-RTOS <https://github.com/Firat17aze/AVR-Zero-Alloc-RTOS>

• Implemented a cooperative real-time scheduler in C for the ATmega328P, developing an embedded RTOS foundation without dynamic memory allocation.

Avr-ring-buffer <https://github.com/Firat17aze/avr-ring-buffer>

• Implemented a circular buffer data structure in C for the ATmega328P to support efficient embedded data storage and retrieval.

Atmega328p-internal-telemetry <https://github.com/Firat17aze/atmega328p-internal-telemetry>

• C system for the ATmega328P that monitors internal silicon temperature and voltage using custom drivers and Kalman filtering.

Fault-Injection-Recovery-Analysis (FIRA) <https://github.com/Firat17aze/Fault-Injection-Recovery-Analysis>

• Shows how embedded systems can survive random faults and crashes, using custom drivers and a watchdog timer to keep everything running smoothly.

UDP-Reliable-Drone <https://github.com/Firat17aze/UDP-reliable-drone>

• Reliable UDP communication protocol for drone command and telemetry with CRC validation and ACK/retry mechanism

SecureBoot-AVR <https://github.com/Firat17aze/SecureBoot-AVR>

• Secure bootloader for ATmega328P with Ed25519 signature verification and anti-rollback protection.

Work Experience

Azersilah CJSC | AI/Embedded Systems Engineer Intern

May 2026 - Aug 2026

• Collaborated with the software team to develop AI models for UAV systems using embedded C++ and Python.
• Built embedded software solutions for integrating AI models on embedded Linux and Zynq UltraScale+ MPSoC platforms.
• Supported the hardware team in designing FPGA-based hardware modules involving signal processing and data structures.

Handshake AI | AI Trainer Fellowship

May 2026 - Aug 2026

• Evaluated and compared AI model responses across text, image, and video tasks, assessing accuracy, reasoning, instruction-following, and overall response quality.

ACCOMPLISHMENTS

• Honors Diploma of High School

• **Dean's List All Time**

CERTIFICATIONS

• Microsoft Certified Azure AI-900
• Microsoft Certified Azure AZ-900
• Codecademy Create Video Games With Phaser.js
• Codecademy Learn The Basics of Android Course

• Microsoft Certified Azure DP-900
• Codecademy Learn C Skill Path
• Codecademy OpenAI Coding With JavaScript

LANGUAGES

• Turkish(C2 Proficient (C2))
• English(C1 Advanced (C1))
• Russian(A2 Elementary (A2))
• Azerbaijani(Native or Bilingual)